AMTI (NMTC) - 2014

BHASKARA CONTEST - FINAL - JUNIOR LEVEL

1. Two circles S_1 and S_2 intersect at points A and B. The tangent at A to S_1 meets S_2 at C and the tangent at A to S_2 meets S_1 at D. A line through A interior to the angle CAD meets S_1 at M and S_2 at N and meets the circumcircle of triangle ACD at P. Prove that AM = NP.

2. Let a, b, c, d be positive real numbers. Show that

$$\frac{ab+bc+ca}{a^3+b^3+c^3}+\frac{ab+bd+da}{a^3+b^3+d^3}+\frac{ac+cd+da}{a^3+c^3+d^3}+\frac{bc+cd+db}{b^3+c^3+d^3}$$
$$\leq \min\left\{\frac{a^2+b^2}{(ab)^{3/2}}+\frac{c^2+d^2}{(cd)^{3/2}}, \frac{a^2+c^2}{(ac)^{3/2}}+\frac{b^2+d^2}{(bd)^{3/2}}, \frac{a^2+d^2}{(ad)^{3/2}}+\frac{b^2+c^2}{(bc)^{3/2}}\right\}.$$

3. Find prime numbers p such that $4p^2 + 1$ and $6p^2 + 1$ are also prime numbers.

4. a) Find all positive integral solutions x, y, z of the equation xy + yz + zx = xyz + 2.

   b) ABC is an equilateral triangle. D is a point inside the triangle such that DA = DB. E is a point that satisfies the two conditions (i) $\angle$DBE = $\angle$DBC and (ii) BE = AB.

5. a) Show that the numbers 1 to 15 cannot be divided into a group A to 2 numbers and a group B of 13 numbers in such a way that the sum of the numbers in B is equal to the product of the numbers in A.

   b) Squares ABCD and BCFG are drawn outside of a triangle ABC. Prove that if DG is parallel to AC then the triangle ABC is isosceles.

6. There are 13 white, 15 black and 17 red beads on a table. You have many number of beads of these colours with you. In one step 2 beads on the table of different colours are closen by you and you replace each one by a bead of the third colour from you. After how many such steps you will have all the beads of the same colour ?

7. a) If a, b, c, d are positive real numbers such that a + b + c + d = 1, show that $\dfrac{a^3}{b+c}+\dfrac{b^3}{c+d}+\dfrac{c^3}{d+a}+\dfrac{d^3}{a+b}\geq\dfrac{1}{8}$.

   b) A 4-digit number n not containing the digit 9 is a square of an integer. If we increase every digit of n by 1 we get a square of another integer again. Find all such n.

8. a) Find all positive real numbers x, y, z which satisfy the following equations simultaneously.
$$\begin{aligned} x^3+y^3+z^3 &= x+y+z \\ x^2+y^2+z^2 &= xyz \end{aligned}$$

   b) Do there exist 10 distinct integers such that the sum of any 9 of them is a perfect square ?

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